Abdalla soliman 2010225594

pilot engine analysis

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Date

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1. File Report

Table 1. File Information for FFF 1

Case	FFF 1
File Path	C:\Users\abd\Desktop\Aero\mesh\12345mesh_files\dp0\FFF-1\Fluent\SYS-3-1-00100.dat.h5
File Date	25 November 2025
File Time	09:32:22 PM
File Type	CFF
File Version	--

2. Mesh Report

Table 2. Mesh Information for FFF 1

Domain	Nodes	Elements
air	78355	304974

3. Physics Report

Table 3. Domain Physics for FFF 1

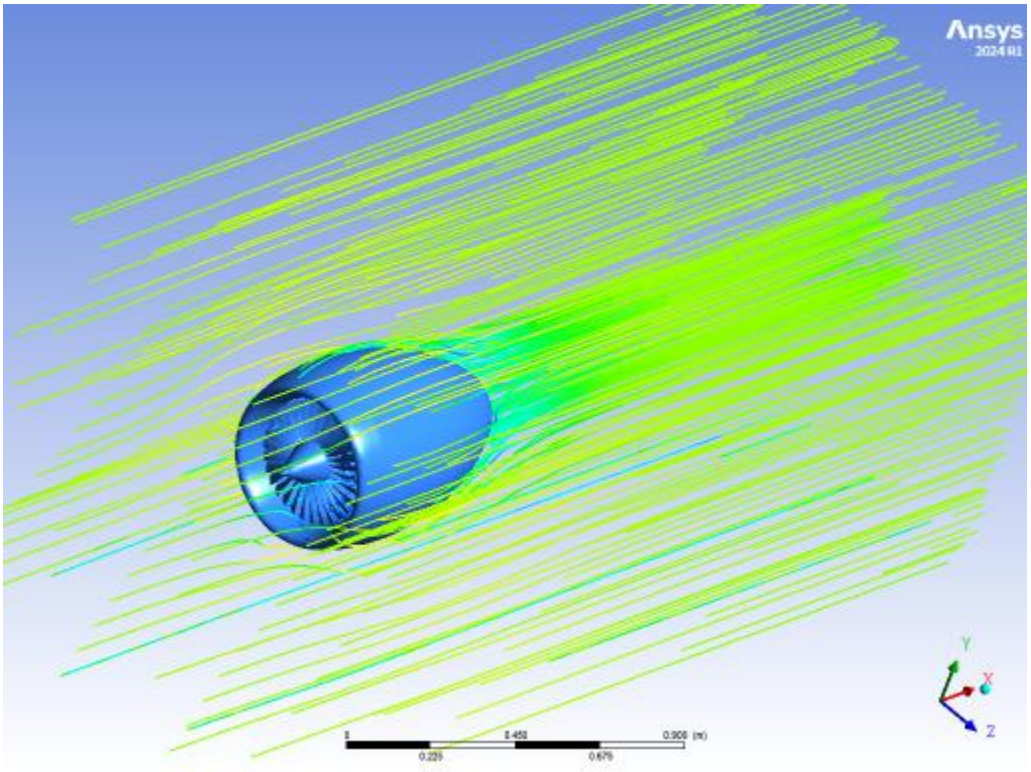
Domain - air	
Type	FLUID
Location	air

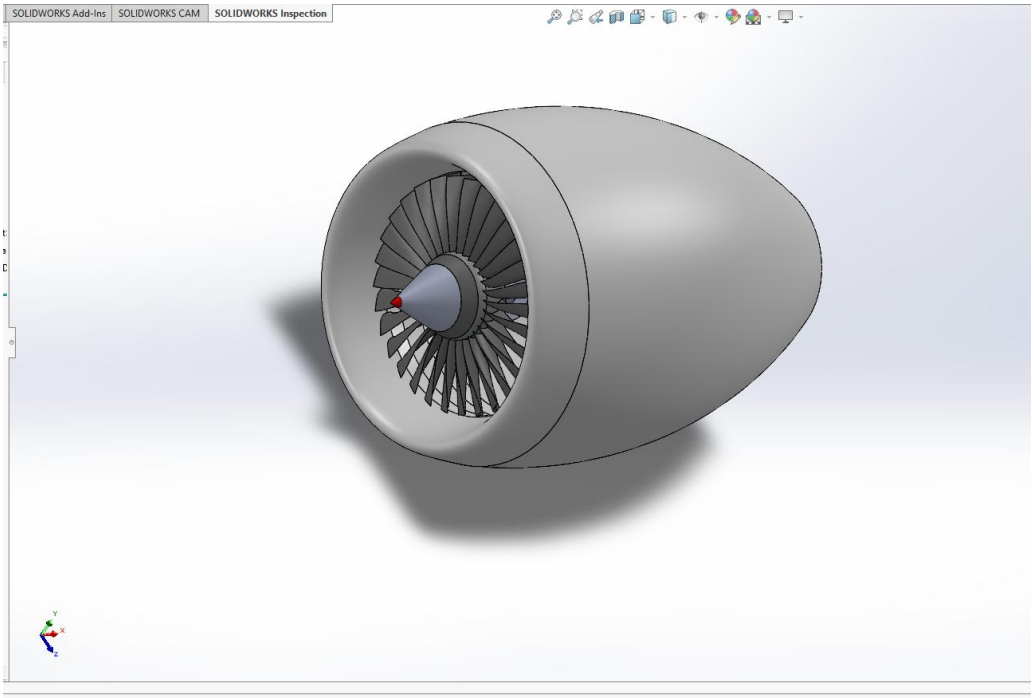
Table 4. Boundary Physics for FFF 1

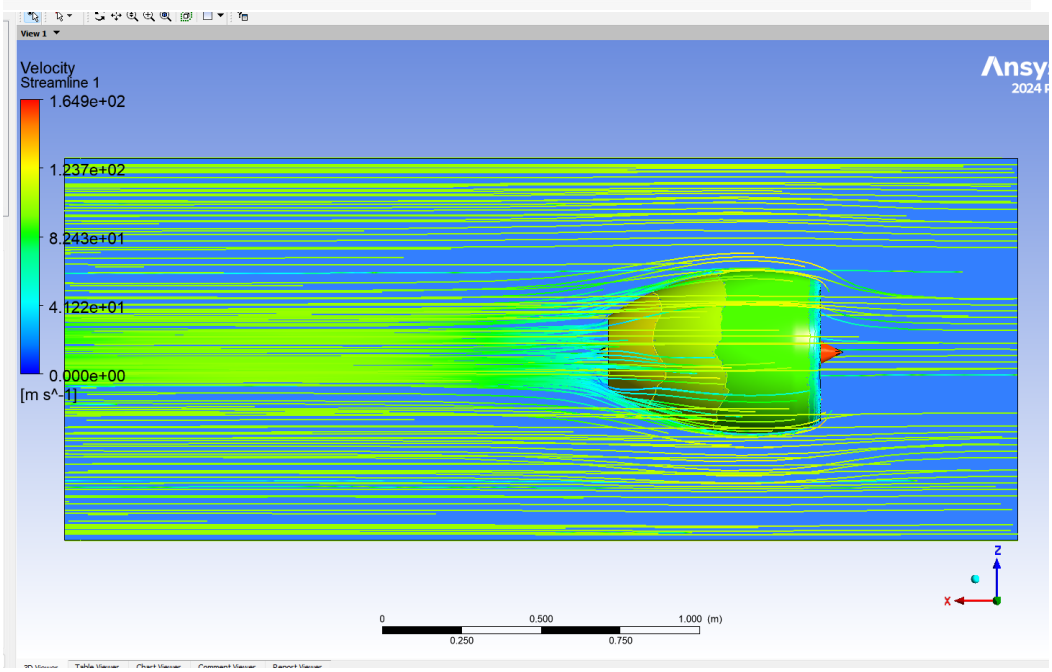
Domain	Boundaries	
air	**Boundary - base**	
	Type	WALL
	Location	base
	Boundary - engine	
	Type	WALL
	Location	engine
	Boundary - inlet	
	Type	VELOCITY-INLET
	Location	inlet
	Boundary - outlet	
	Type	PRESSURE-OUTLET
	Location	outlet
	Boundary - wall	
	Type	SYMMETRY
	Location	wall

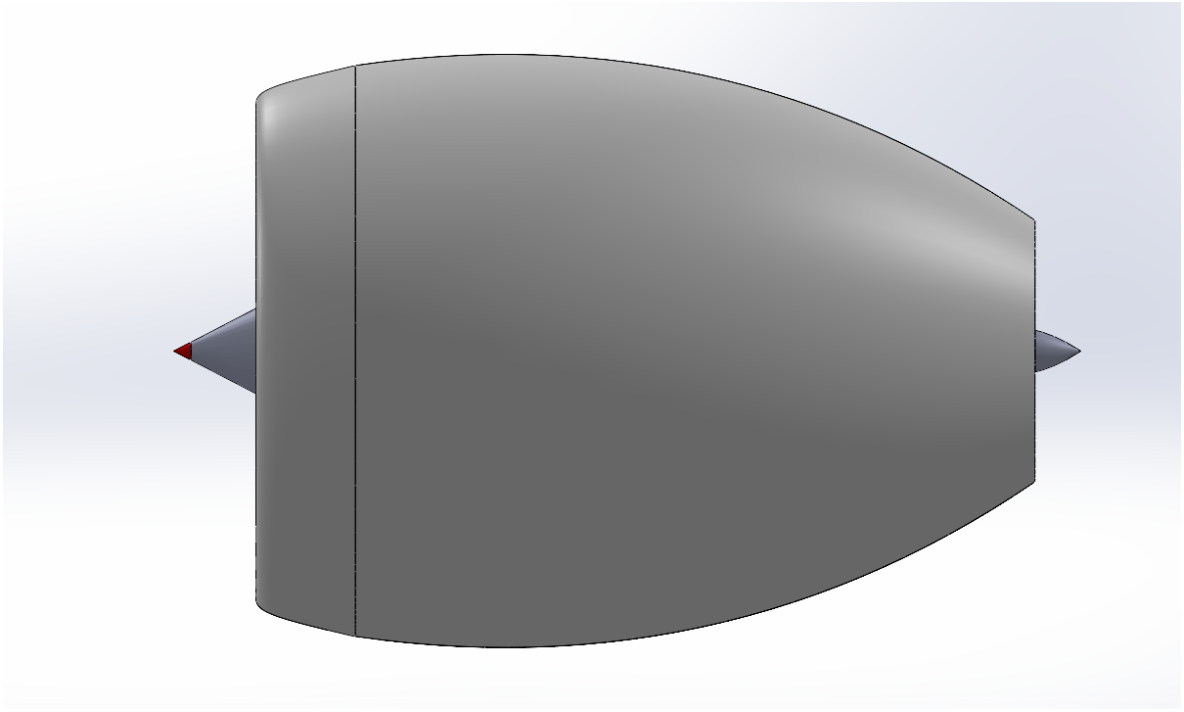
4. User Data

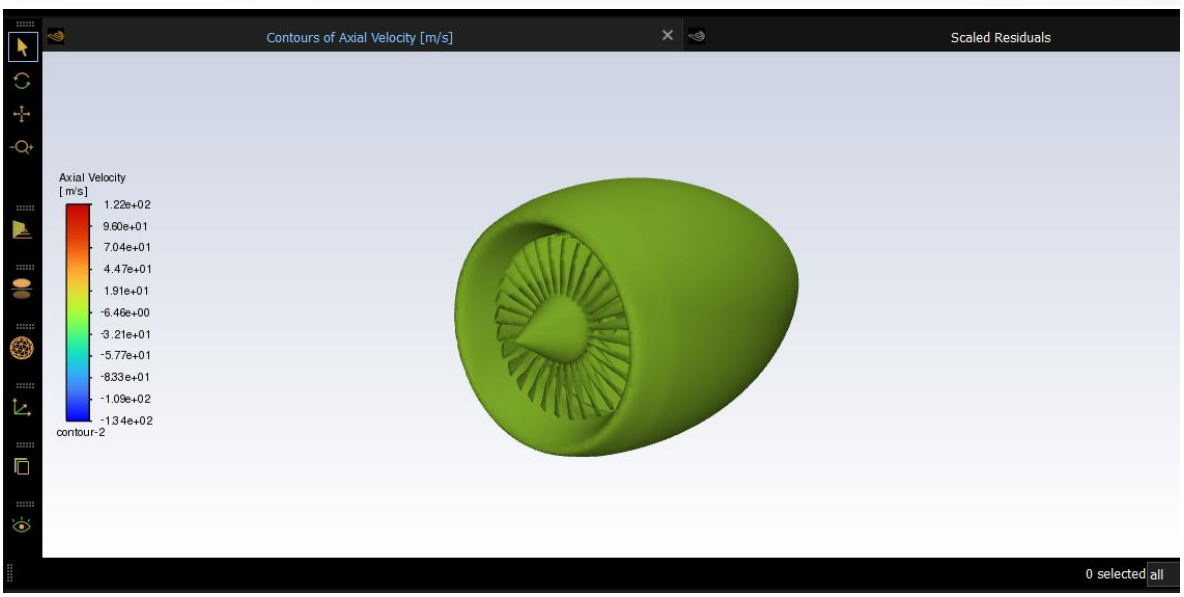
Figure 1.

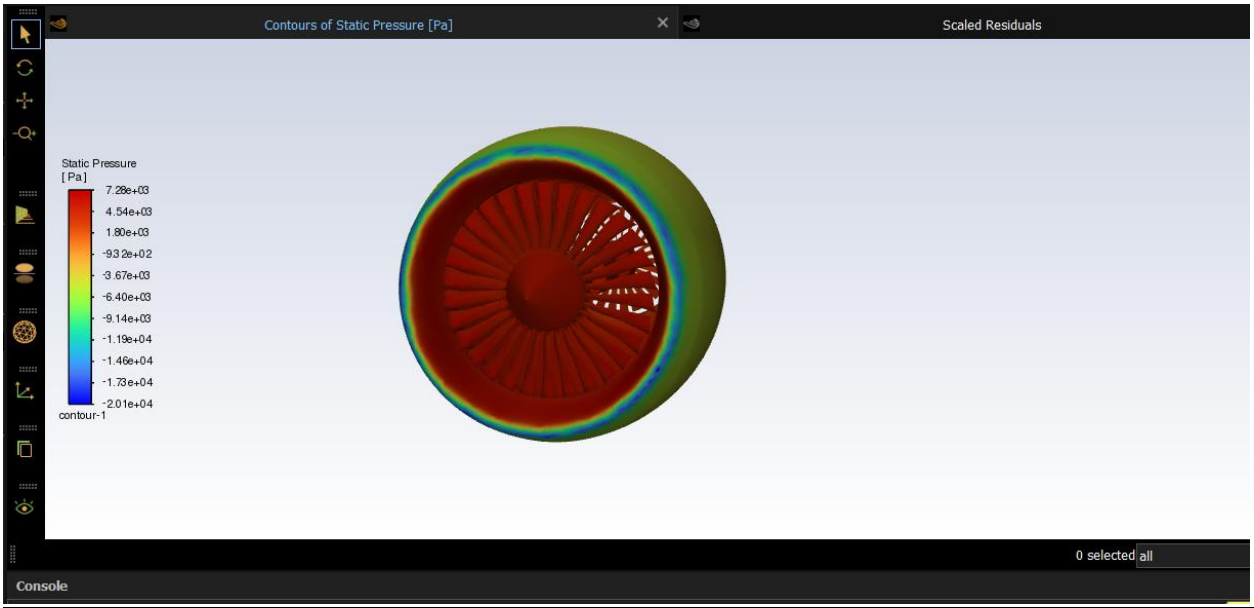


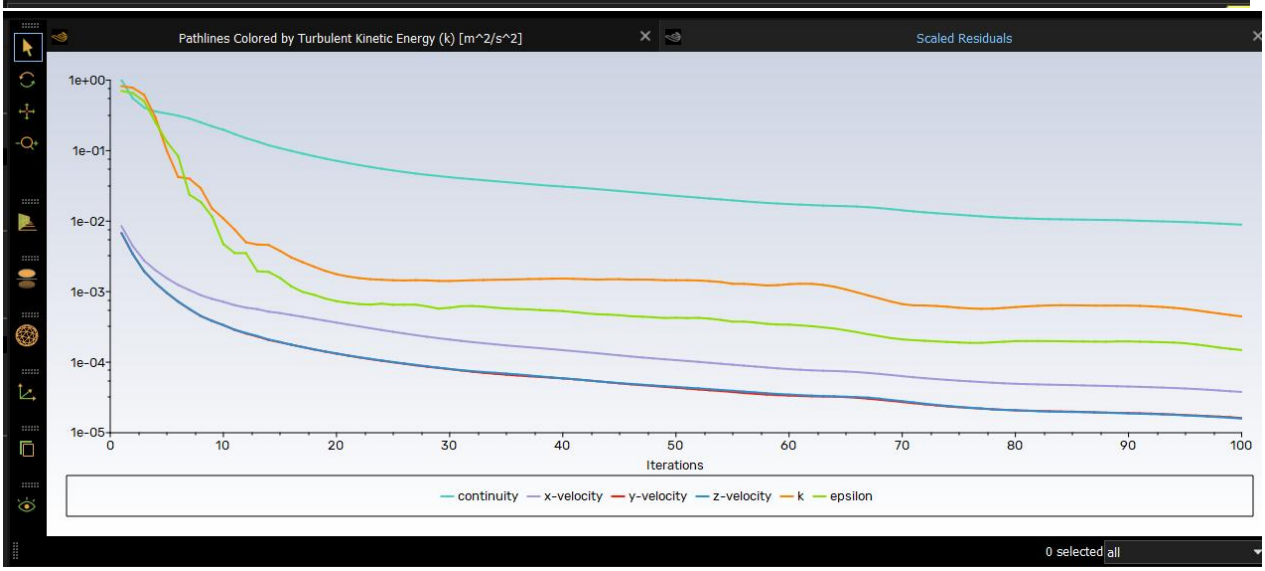

SOLIDWORKS Add-Ins
SOLIDWORKS CAM
SOLIDWORKS Inspection

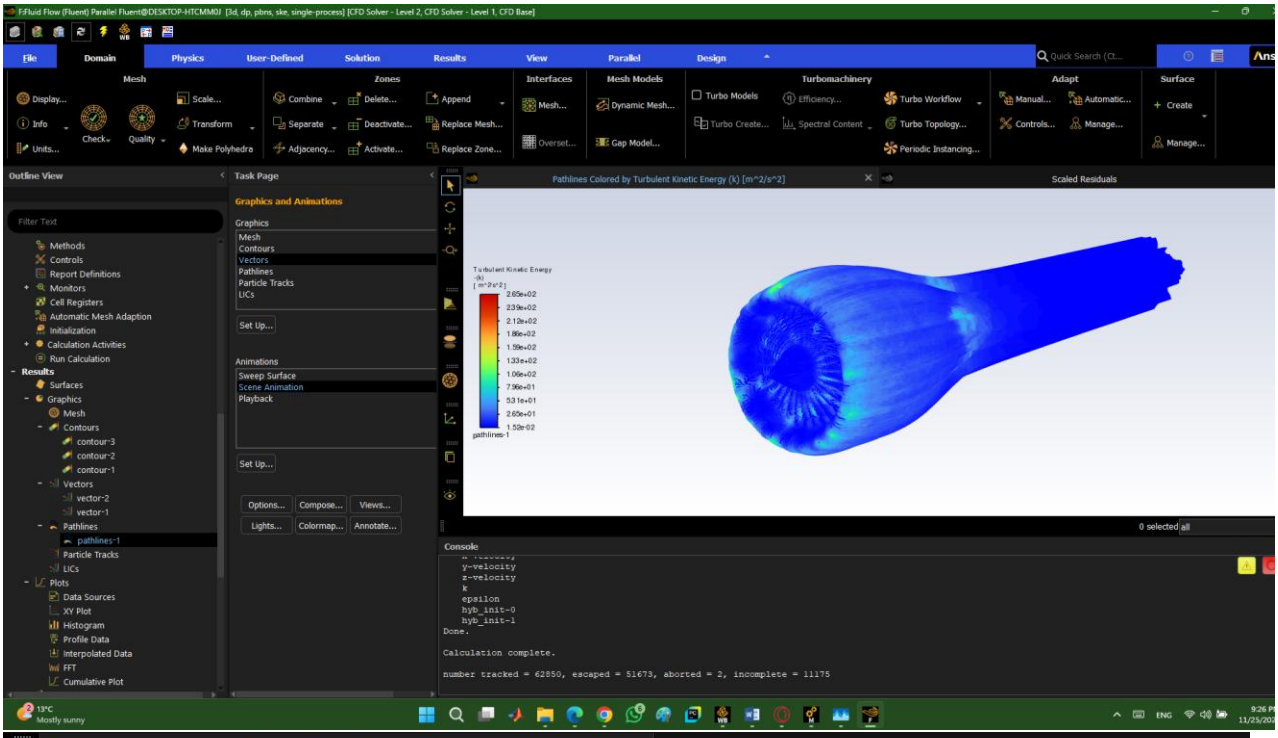

View 1
Velocity
Streamline 1
1.649e+02
1.237e+02
8.243e+01
4.122e+01
0.000e+00
[m s^-1]
Ansys
2024
0
0.500
1.000 (m)
0.250
0.750
Z
X

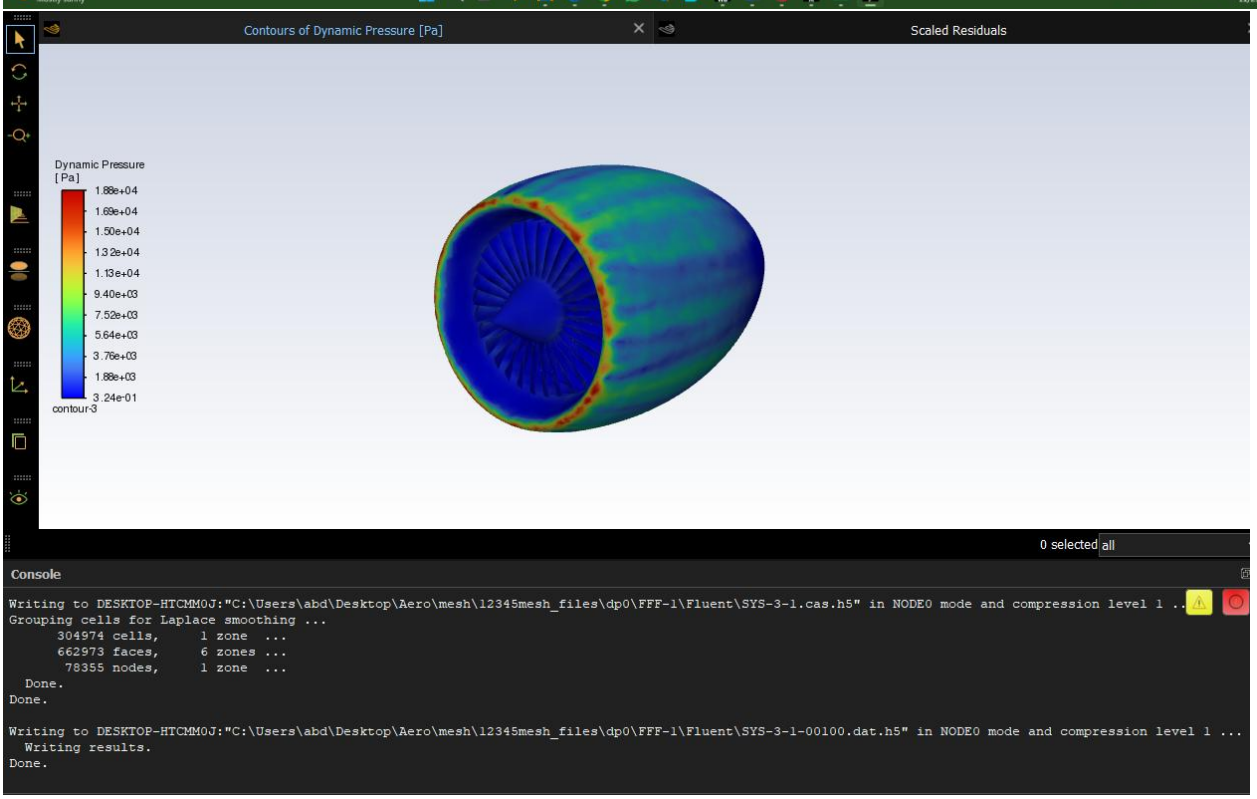


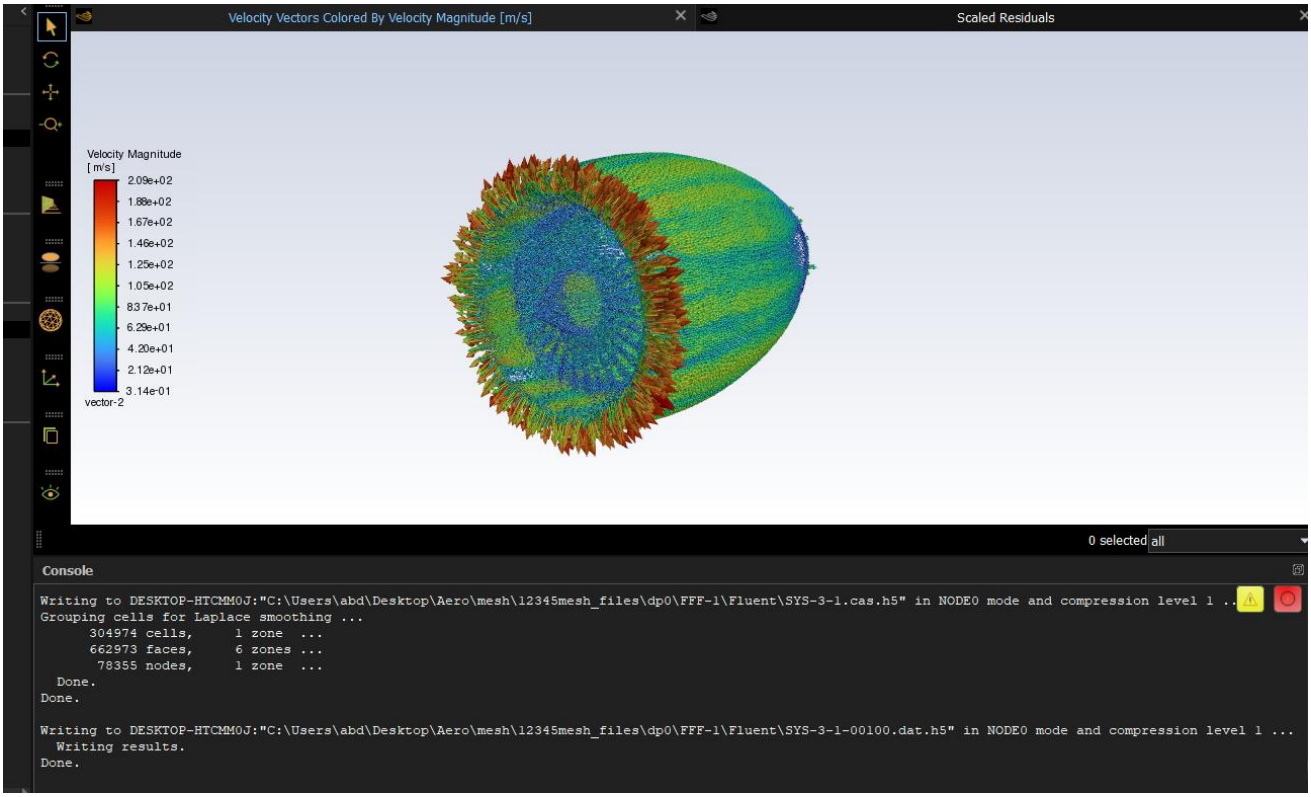

Contours of Axial Velocity [m/s]
Scaled Residuals
Axial Velocity
[m/s]
1.22e+02
9.60e+01
7.04e+01
4.47e+01
1.91e+01
-6.46e+00
-3.21e+01
-5.77e+01
-8.33e+01
-1.09e+02
-1.34e+02
contour-2
0 selected
all


Contours of Static Pressure [Pa]
Scaled Residuals
Static Pressure
[Pa]
7.28e+03
4.54e+03
1.80e+03
-9.32e+02
-3.67e+03
-6.40e+03
-9.14e+03
-1.19e+04
-1.46e+04
-1.73e+04
-2.01e+04
contour-1
0 selected all
Console


Pathlines Colored by Turbulent Kinetic Energy (k) [m^2/s^2]
Scaled Residuals
1e+00
1e-01
1e-02
1e-03
1e-04
1e-05
0
10
20
30
40
50
60
70
80
90
100
Iterations
continuity
x-velocity
y-velocity
z-velocity
k
epsilon
0 selected all


Pathlines Colored by Turbulent Kinetic Energy (k) [m^2/s^2]
Scaled Residuals
Console
Calculation complete.
number tracked = 62850, escaped = 51673, aborted = 2, incomplete = 11175


Contours of Dynamic Pressure [Pa]
Scaled Residuals
Dynamic Pressure
[Pa]
1.88e+04
1.69e+04
1.50e+04
1.32e+04
1.13e+04
9.40e+03
7.52e+03
5.64e+03
3.76e+03
1.88e+03
3.24e-01
contour-3
0 selected all
Console
Writing to DESKTOP-HTCMM0J:"C:\Users\abd\Desktop\Aero\mesh\12345mesh_files\dp0\FFF-1\Fluent\SYS-3-1.cas.h5" in NODE0 mode and compression level 1 ...
Grouping cells for Laplace smoothing ...
304974 cells, 1 zone ...
662973 faces, 6 zones ...
78355 nodes, 1 zone ...
Done.
Done.
Writing to DESKTOP-HTCMM0J:"C:\Users\abd\Desktop\Aero\mesh\12345mesh_files\dp0\FFF-1\Fluent\SYS-3-1-00100.dat.h5" in NODE0 mode and compression level 1 ...
Writing results.
Done.


Velocity Vectors Colored By Velocity Magnitude [m/s]
Scaled Residuals
Velocity Magnitude
[m/s]
2.09e+02
1.88e+02
1.67e+02
1.46e+02
1.25e+02
1.05e+02
8.37e+01
6.29e+01
4.20e+01
2.12e+01
3.14e-01
vector-2
0 selected
all
Console
Writing to DESKTOP-HTCMM0J:"C:\Users\abd\Desktop\Aero\mesh\12345mesh_files\dp0\FFF-1\Fluent\SYS-3-1.cas.h5" in NODE0 mode and compression level 1 ...
Grouping cells for Laplace smoothing ...
304974 cells, 1 zone ...
662973 faces, 6 zones ...
78355 nodes, 1 zone ...
Done.
Done.
Writing to DESKTOP-HTCMM0J:"C:\Users\abd\Desktop\Aero\mesh\12345mesh_files\dp0\FFF-1\Fluent\SYS-3-1-00100.dat.h5" in NODE0 mode and compression level 1 ...
Writing results.
Done.